

Checkmates and Incentives

Time Pressure, Position, and Gender in Elite Chess Errors

A two-cycle replication: FIDE Candidates **2024** & **2026**

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Roadmap

The question

Where this fits

Data: two cycles

Measuring an “error”

Empirical strategy

Findings I: within a cycle

An honest detour

Findings II: pooled across cycles

Findings III: a second outcome & the same people

Is it defensible?

Limitations & future

The question

The big picture (in plain words)

Even the best players in the world make mistakes. We ask:

Research question

What makes an elite player more likely to err on a given move, and do **men and women respond differently** to the same pressures?

Three pressures:

- **Time pressure**, a ticking clock.
- **Position**, winning, equal, or losing?
- **Game phase**, opening, middlegame, endgame.

A clean, high-stakes laboratory for *decision-making under pressure*: world-class players, enormous incentives, and **every move scored objectively by an engine**.

Why this setting is special, and now, repeatable

- **World elite, identical conditions.** The Candidates decides the World-Championship challenger, maximum stakes, controlled environment.
- **An objective referee.** A chess engine scores every position, so “mistake” is *measured*, not judged.
- **Two parallel sections.** Open (men) and Women’s, same format, a natural side-by-side gender comparison.
- **Two cycles run the same way.** 2024 (Toronto) and **2026 (Cyprus)** are the only cycles with both sections as identical double round-robins, **same time control** ⇒ a built-in **replication test**.

The arc of this talk

1. **2024** gave a sharp result: women err more **specifically when winning**.
2. **2026** is an *independent* cohort under the same rules, does it replicate?
3. A twist: 2026 *alone* first looked like it **broke** the result...
4. ...but pooling both cycles (the right tool) shows it **holds**, and a **second outcome** (severity) and the **8 repeat players** reinforce it.

The honest through-line

A finding you only believe *after* you tried to break it, and after you caught yourself nearly mis-reading one cycle in isolation.

Where this fits

We are not the first to ask, here is what is known

A real economics literature uses chess as a clean lab for **gender, competition, and performance**, mostly on *mass* datasets of ordinary tournaments.

- Backus, Cubel, Guid, Sánchez-Pagés & Mañas (2023, *Quant. Econ.*): women underperform men of **equal ability**; the effect is driven by the **opponent's gender** (women play worse *against men*), not by men playing better.
- de Sousa & Hollard / Smerdon et al. (perf. under time constraint, ~1.8M games): the female disadvantage **grows as the time control tightens** (faster chess).
- Dreber, Gerdes & Gränsmark (“Chess girls don’t cry”, 2022, *JEBO*): gender *composition* of a game shifts effort and risk-taking among the super-elite.

The shared shape of prior work

Outcome = **game result or aggregate play quality**; mechanism = **opponent gender** or **speed**; identification leans on **rating to proxy ability**.

What we do differently (our contribution)

1. **Move-level, state-conditional.** We score *every half-move* and condition on **board state** (winning / equal / losing). The gap we find is **specific to winning positions**, a *conversion* failure, invisible to game-result studies.
2. **We flip the time-pressure story.** In our elite data the time-pressure gap is **null**; the action is in *position*, not the clock, the opposite emphasis to the “faster chess hurts women more” result.
3. **Parallel same-tournament design.** Open vs Women’s Candidates, **identical rules, format, stakes, dates**, a cleaner side-by-side than pooling heterogeneous opens.
4. **Built-in replication & within-person evidence.** **Two cycles**, player fixed effects, **8 repeat players**, and a wild cluster bootstrap, rare in this literature, which is usually a single cross-section.

Scope, stated honestly

Sections are gender-segregated. so we *cannot* test the prior literature’s

Data: two cycles

Two Candidates cycles, one design

	2024	2026
Venue	Toronto, Canada	Pegeia, Cyprus
Sections	Open + Women's	Open + Women's
Format	8-player double RR	8-player double RR
Time control	120/40 (+30m, +30s $\geq$ 41)	W: 90/40 (+30m, +30s)
Open Elo range	2632–2803	2698–2810
Women Elo range	2432–2553	2470–2578
Games	112	112
Moves (modelled)	10,656 (10,480)	10,690 (10,574)
Errors (CPL $\geq$ 50)	627	684

Source: official **Lichess broadcast PGNs**, ~100% engine-eval + clock coverage. **Pooled**: 224 games, **21,054** move-player observations, **24 distinct players**, **8 of them play in both cycles**.

What one annotated move looks like

8. Na3?!

```
{ [%eval 0.09] [%clk 1:11:21] }
```

Read as: White's 8th move was an inaccuracy; the engine now rates the position **+0.09 pawns**; White has **1h 11m 21s** left on the clock.

- [%eval] → position quality.
- [%clk] → time remaining.

From two PGN files per cycle we build **every move, with its eval and clock**, no hidden data, fully reproducible.

Same pipeline, both cycles

2026's time control is **identical to 2024's**, so the *exact same* feature-engineering code produces both datasets, the only thing that changes is the games.

The Elo catch, and how the repeat players help

- Within each cycle, **men** and **women do not overlap** in rating (~ 80 -point gap). So “gender” and “skill” are entangled in any single section.
- **New leverage in 2026: 8 players appear in both cycles**, 3 men (Nakamura, Caruana, Praggnanandhaa) and 5 women (Goryachkina, A. Muzychuk, Tan, Vaishali, Lagno).

Why this matters

Linking players across cycles by **FIDE id** lets us hold *the same person* fixed and watch them twice, the cleanest skill-vs-gender check one tournament can't give.

Measuring an “error”

The outcome: centipawn loss (CPL)

Definition

100 cp = 1 pawn. **CPL = advantage you gave up** versus the engine's best move.
Position +150 before, +120 after $\Rightarrow$ CPL = 30.

CPL	Label
50–99	inaccuracy
100–199	mistake
200+	blunder

- Primary outcome: **error = 1 if CPL $\geq$ 50**.
- Second outcome (later): **CPL magnitude**, how *bad* the slip is.
- Standard Lichess cutoffs $\Rightarrow$ no “researcher’s choice”.
- We use the *relative drop* between positions, cancelling engine offsets.

Two measurement choices we got right (both cycles)

1. Forced mate (#)

Engines print “#” for forced mate. Naively these get *dropped*, losing the **most decisive blunders**. Fix: treat mate as ± 100 pawns, **cap CPL at 1000**.

2. The clock / time control

- **+30-minute bonus credited at move 40** (verified against the clock jump).
- **+30s increment**: every move for Women (90+30); from move 41 for Men (120/40).
- **2026 uses the identical control**, confirmed from the broadcast headers, so the same logic applies unchanged.

Empirical strategy

How we estimate the effects

Linear probability model for an error on a move:

$$\text{error} = \beta_1 \text{ time left} + \beta_2 \text{ position} + \dots + \gamma (\text{female} \times \text{pressure}) + \varepsilon$$

- **Spec 1 (no player effects):** gender + Elo + controls + interactions.
- **Spec 2 (player fixed effects):** each player vs. their own average, absorbs all fixed skill/style; **gender×pressure interactions stay identified.**

The pooled upgrade (the key new step)

Stack both cycles. Add a **cycle effect** and let gender effects **vary by cycle**. Link players across years by **FIDE id**, cluster on **person** ($\Rightarrow$ **24 clusters**, up from 16). Gender effects can now be tested for **stability across cycles**.

Findings I: within a cycle

The non-gender effects are rock-solid in both cycles

Effect on P(error)	2024	2026	Pooled
Time remaining (full→empty)	−0.110***	−0.100***	−0.106***
Position: winning (vs equal)	+0.049***	+0.088***	+0.066***
Position: losing (vs equal)	+0.162***	+0.218***	+0.189***
Phase: mid (vs early)	−0.023***	−0.024***	−0.023***

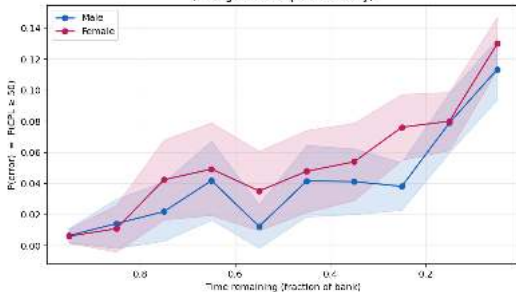
Time pressure raises errors; losing positions are far more error-prone; winning positions modestly more.

Same story, same magnitudes, both cycles. (***) $p < 0.01$

Time pressure bites, both cycles

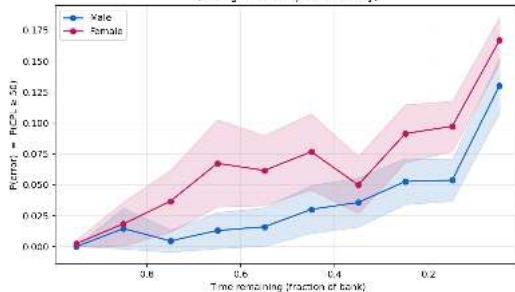
2024

Error probability rises sharply as time runs low
(both genders respond similarly)



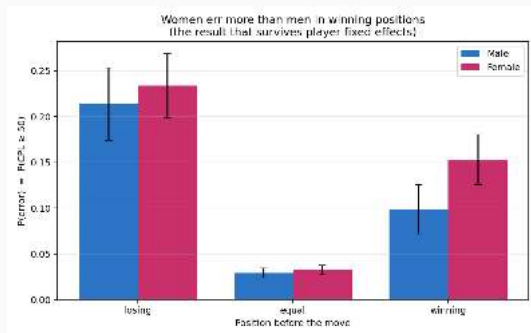
2026

Error probability rises sharply as time runs low
(both genders respond similarly)



Error probability climbs steeply as the clock drains, for everyone. The lines largely track each other, **time pressure is not where a robust gender gap lives.**

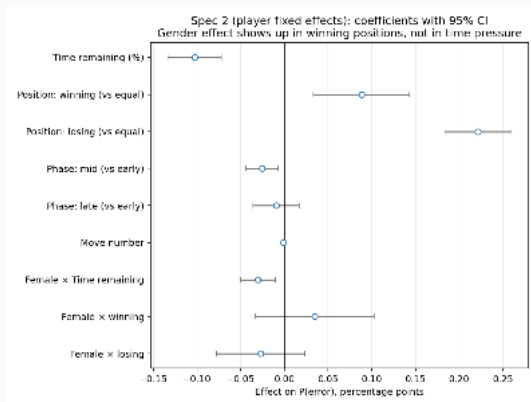
2024's sharp result: the gap is in *winning* positions



- **Losing:** errors explode for both (~1 in 5).
- **Equal:** both very accurate (~3%).
- **Winning:** women 15.2% vs men 9.9%.

The only place gender shows up is **when ahead**, and it **survived player fixed effects, a bootstrap, and a logit** (2024 paper).

2026 alone: a worrying first look



Looking at **2026** by itself:

- Female×winning: **not significant** in the single-cycle fit.
- Female×time: *appeared* significant and negative.

First impression: **“2024 didn’t replicate.”**
...this is exactly where one can fool oneself. Hold that thought.

An honest detour

A cautionary tale (and why pooling is the fix)

What nearly fooled us

A *single cycle* has only 16 players $\Rightarrow$ 16 clusters. Cluster-robust p-values over-reject and individual estimates are **specification-sensitive**. The 2026-alone “time” effect and “no winning effect” were artifacts of that fragility.

The right tool

Don't compare two noisy tables by eye, **pool both cycles into one model**, let the gender effects differ by cycle, and **formally test** whether they do. With the 8 repeat players and person clustering, we get 24 clusters and real power.

The lesson generalizes: replication is a pooled-estimation question, not a “count the stars in each table” question.

Findings II: pooled across cycles

The pooled model (gender effects split by cycle)

Term	Spec A (no FE)	Spec B (person FE)
Time remaining (%)	−0.106***	−0.108***
Position: winning	+0.066***	+0.067***
Position: losing	+0.189***	+0.188***
Female × winning, 2024	+0.036	+0.032
Female × winning, 2026	+0.058**	+0.059**
Female × time, 2024	+0.001	−0.006
Female × time, 2026	−0.020*	−0.016
Cycle 2026 (vs 2024)	−0.003	−0.003

$N = 21,054$ move-obs; 24 person clusters; 8 players in both cycles. The winning-position gap is **positive in both cycles** and significant in 2026 *within the better-powered pooled model*.

(* $p < .10$, ** $p < .05$, *** $p < .01$)

Does it replicate? Formal tests + descriptives

Equality test: is the 2024 effect = 2026 effect?

Gender effect	diff (24–26)	<i>p</i>
Female × winning	−0.022	0.58
Female × time	+0.022	0.15
Female × losing	−0.016	0.43

None rejected ⇒ the cycles are statistically *indistinguishable*. The winning gap **replicates**.

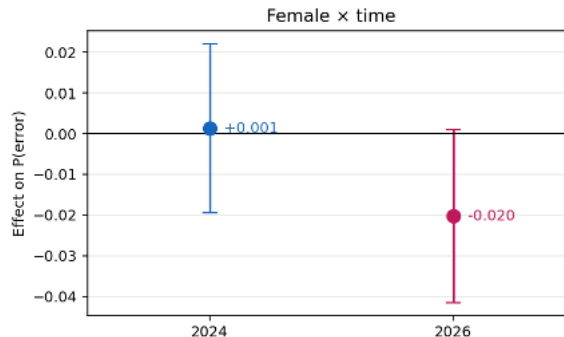
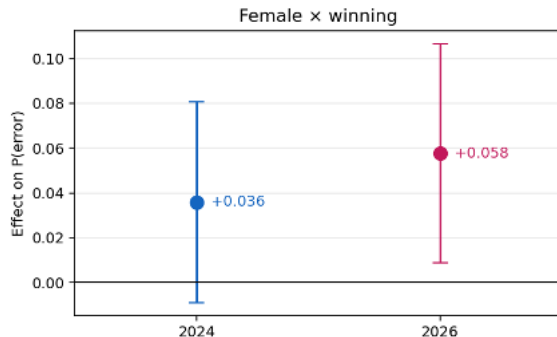
P(error) in winning positions

	Male	Female
2024	9.9%	15.2%
2026	11.6%	18.7%

Women ~5–7 pp higher *in both cycles*.

The picture: gender effects by cycle (95% CI)

Gender interaction effects by cycle (pooled model, 95% CI)
2024's winning-position gap does not carry to 2026; a time-pressure gap appears



Female x winning sits above zero in *both* cycles; **Female x time** straddles zero, the fragile signal.

Stress test: does the pooled gap survive proper inference?

Same battery the 2024 headline had to pass, now on the pooled model.

Test	Estimate	Verdict
Female×winning (pooled) WCB	+0.048, $p = \mathbf{0.013}$	survives
with person fixed effects	$p = \mathbf{0.020}$	survives
Replication (2024 = 2026) WCB	$p = 0.58$	not rejected
Female×winning, 2026 alone WCB	$p = 0.036$	holds
Female×time (pooled) WCB	$p = 0.35$	null (fragile)
Logit (sign check)	winning > 0 both cycles	consistent

Wild cluster bootstrap (Rademacher, $B = 1999$, clustered on person), the recognized fix for few clusters. **The winning-position gap is credible; the time signal is not.**

Findings III: a second outcome & the same people

Severity: not just *whether*, but *how much*

Expected centipawns lost per move	
Term	coef (cp)
Position: winning	+15.7***
Position: losing	+33.6***
Female×winning 2024	+7.2*
Female×winning 2026	+18.9***
Female×winning (pooled)	+13.4***

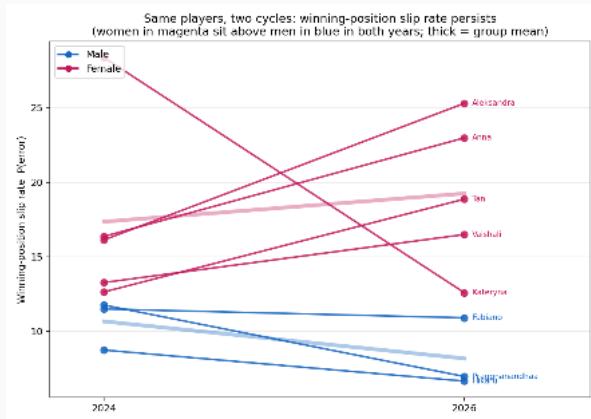
WCB $p = 0.009$.

The winning-position gap appears in **both** outcomes, incidence and magnitude.

Mean CPL of a winning slip		
	Male	Female
2024	161	180
2026	148	212

Within winning-slips: female +39.8 cp
($p = 0.046$). **More often and more severe.**

The 8 repeat players, same people, two cycles

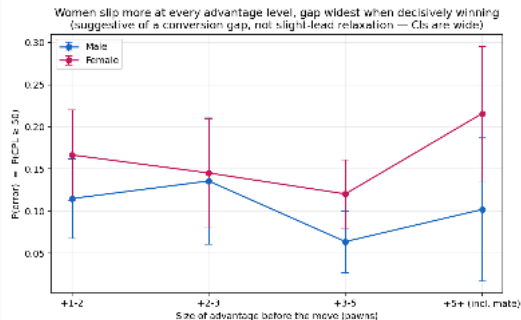


- Group means (winning $P(\text{error})$):
Men 10.7% $\rightarrow$ 8.2%; Women 17.4% $\rightarrow$ 19.2%.
- **All 5 repeat women** sit above **all 3 repeat men**, in both cycles.

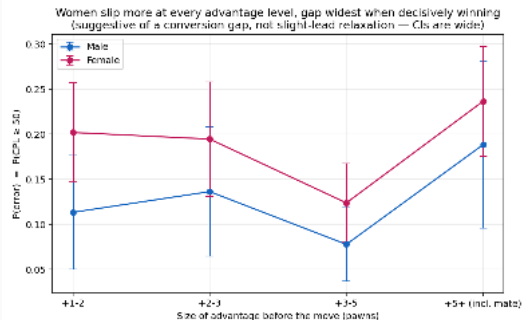
Honest nuance: *individual* year-to-year persistence is weak ($\text{corr} \approx +0.24$; Lagno swung 28% $\rightarrow$ 13%). It's the **group ordering**, holding the people fixed, that is striking, a vivid vignette, not a powered test ($n = 3, 5$).

Mechanism: where inside “winning”? (both cycles)

2024



2026



Women slip more at *most* advantage levels, gap often widest when decisively winning, consistent with weaker **conversion of won positions** rather than slight-lead relaxation. (Within-winning sub-splits are small $\Rightarrow$ suggestive.)

Is it defensible?

Why you can trust the two-cycle result

- **Two independent cohorts.** A different field, a year apart, same rules, and the winning-position gap recurs.
- **Survives the proper small-sample test.** Wild cluster bootstrap on the pooled model: $p = 0.013$ (incidence), $p = 0.009$ (severity).
- **Two outcomes agree.** Both “does a slip happen” and “how big is it” point the same way.
- **Within-person evidence.** Holding the same 8 people fixed across cycles preserves the gender ordering.
- **Reproducible.** Two PGN files per cycle + code → every table and figure.

Limitations & future

Honest limitations

- **Still no Elo overlap within a cycle.** Gender and skill can't be *fully* separated at the field level; the 8 repeat players help but don't create overlap.
- **Still few clusters.** 24 beats 16, but inference leans on the bootstrap; individual per-cycle estimates remain spec-sensitive.
- **Two cycles is better than one, not “many.”** The direction is encouraging; a third cycle would harden it further.
- **Conditional-severity cells are small** ($n = 336$ winning slips), that piece is the soft part; the per-move expected-CPL result is the robust one.
- **Engine evals as ground truth** (standard; we use relative drops).

Where we go next

1. **A third cycle** when available, push from “replicates once” toward a trend.
2. **Mechanism with power**, pooled within-winning model (conversion vs. relaxation), now that two cohorts give more winning-position moves.
3. **Person-level trajectories**, track the repeat players as more cycles accrue.
4. **Domain validation**, take “weaker conversion under pressure” to a coach / GM.

Already done since 2024

Replication (2026), **severity outcome**, **pooled bootstrap**, **repeat-player check**.

What we found

- Time pressure and bad positions drive errors, **equally by gender**.
- **Women give back won positions more than men, in both 2024 and 2026**, in **both incidence and severity**, surviving the bootstrap.
- The 2026-alone “time” effect was a fragile artifact; pooling corrected it.

Why it's credible

Two independent cohorts, a pooled bootstrap, two agreeing outcomes, within-person evidence, and a fully reproducible pipeline.

Code & data: github.com/xkanav/Checkmates-and-Incentives

Thank you, questions & discussion

Appendix

Appendix, pooled regression, full terms

Term	Spec A	Spec B (FE)
Female (vs Male)	+0.015	,
Female $\times$ cycle2026	+0.021	,
Cycle 2026	−0.003	−0.003
Time remaining (%)	−0.106***	−0.108***
Position: winning	+0.066***	+0.067***
Position: losing	+0.189***	+0.188***
Phase: mid	−0.023***	−0.025***
Phase: late	−0.017*	−0.019*
Move number	−0.0008***	−0.0008***
Female $\times$ winning 2024	+0.036	+0.032
Female $\times$ winning 2026	+0.058**	+0.059**
Female $\times$ time 2024	+0.001	−0.006
Female $\times$ time 2026	−0.020*	−0.016
Female $\times$ losing 2024	−0.009	−0.012
Female $\times$ losing 2026	+0.007	+0.010
<i>N</i> / clusters	21,054 / 24	21,054 / 24
<i>R</i> ²	0.094	0.096

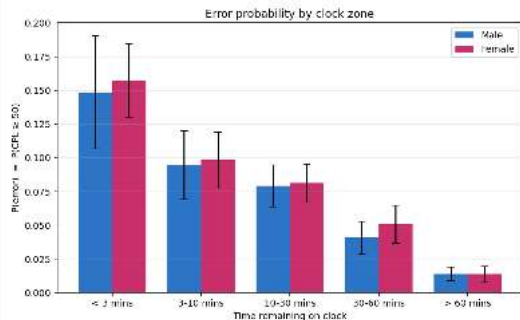
Appendix, repeat players, per person

Player	Sex	winning P(error)		winning mean CPL	
		2024	2026	2024	2026
Goryachkina	F	16.1%	25.3%	17	57
A. Muzychuk	F	16.4%	23.0%	47	99
Tan Zhongyi	F	12.6%	18.9%	22	50
Vaishali	F	13.3%	16.5%	47	45
Lagno	F	28.4%	12.6%	45	31
Caruana	M	11.5%	10.9%	26	25
Praggnanandhaa	M	11.8%	7.0%	21	25
Nakamura	M	8.8%	6.7%	27	18

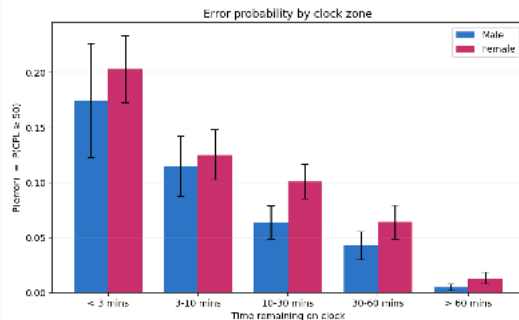
Every woman above every man in both cycles; individual persistence noisy (e.g. Lagno).

Appendix, error by clock zone (both cycles)

2024



2026



Appendix, the model, formally

Pooled, gender effects split by cycle $c \in \{24, 26\}$:

$$\text{error}_{it} = \mu_i + \theta \text{time\%}_{it} + \phi \text{pos}_{it} + \sum_c \gamma_c^w (\text{F} \times \text{win} \times c)_{it} + \sum_c \gamma_c^t (\text{F} \times \text{time} \times c)_{it} + \lambda c_t + \varepsilon_{it}$$

- $i = \text{person (FIDE id)}$, $t = \text{move}$. $\mu_i = \text{person fixed effect}$ (spans both cycles).
- SEs clustered on person (24 clusters); inference via **wild cluster bootstrap**.
- Replication test: $H_0 : \gamma_{24}^w = \gamma_{26}^w$ (and similarly for time).
- Second outcome: replace error with **CPL** (cap 1000) for the severity model.